Assignment 1 — Exercises

0.1

In each of the following situations, indicate whether $f = O(g)$, or $f = \Omega(g)$, or both (in which case $f = \Theta(g)$).

	$f(n)$	$g(n)$
(a)	$n - 100$	$n - 200$
(b)	$n^{1/2}$	$n^{2/3}$
(c)	$100n + \log n$	$n + (\log n)^2$
(d)	$n \log n$	$10n \log 10n$
(e)	$\log 2n$	$\log 3n$
(f)	$10 \log n$	$\log(n^2)$
(g)	$n^{1.01}$	$n \log^2 n$
(h)	$n^2 / \log n$	$n(\log n)^2$
(i)	$n^{0.1}$	$(\log n)^{10}$
(j)	$(\log n)^{\log n}$	$n / \log n$
(k)	$\sqrt{n}$	$(\log n)^3$
(l)	$n^{1/2}$	$5^{\log_2 n}$
(m)	$n2^n$	3^n
(n)	2^n	2^{n+1}
(o)	$n!$	2^n
(p)	$(\log n)^{\log n}$	$2^{(\log_2 n)^2}$
(q)	$\sum_{i=1}^{n} i^k$	n^{k+1}

0.2

Show that, if c is a positive real number, then $g(n) = 1 + c + c^2 + \cdots + c^n$ is:

(a) $\Theta(1)$ if $c < 1$.

(b) $\Theta(n)$ if $c = 1$.

(c) $\Theta(c^n)$ if $c > 1$.

The moral: in big-Θ terms, the sum of a geometric series is simply the first term if the series is strictly decreasing, the last term if the series is strictly increasing, or the number of terms if the series is unchanging.

0.3

The Fibonacci numbers $F_0, F_1, F_2, \ldots$ are defined by the rule

$$F_0 = 0, \quad F_1 = 1, \quad F_n = F_{n-1} + F_{n-2}.$$

In this problem we will confirm that this sequence grows exponentially fast and obtain some bounds on its growth.

(a) Use induction to prove that $F_n \geq 2^{0.5n}$ for $n \geq 6$.

(b) Find a constant $c < 1$ such that $F_n \leq 2^{cn}$ for all $n \geq 0$. Show that your answer is correct.

(c) What is the largest c you can find for which $F_n = \Omega(2^{cn})$?

1.1

Show that in any base $b \geq 2$, the sum of any three single-digit numbers is at most two digits long.

1.2

Show that any binary integer is at most four times as long as the corresponding decimal integer. For very large numbers, what is the ratio of these two lengths, approximately?

1.3

A d-ary tree is a rooted tree in which each node has at most d children. Show that any d-ary tree with n nodes must have a depth of $\Omega(\log n / \log d)$. Can you give a precise formula for the minimum depth it could possibly have?

1.4

Show that

$$\log(n!) = \Theta(n \log n).$$

(*Hint*: To show an upper bound, compare $n!$ with n^n. To show a lower bound, compare it with $(n/2)^{n/2}$.)

1.5

Unlike a decreasing geometric series, the sum of the harmonic series $1, 1/2, 1/3, 1/4, 1/5, \ldots$ diverges; that is,

$$\sum_{i=1}^{\infty} \frac{1}{i} = \infty.$$

It turns out that, for large n, the sum of the first n terms of this series can be well approximated as

$$\sum_{i=1}^{n} \frac{1}{i} \approx \ln n + \gamma,$$

where ln is natural logarithm (log base $e = 2.718\ldots$) and γ is a particular constant $0.57721\ldots$. Show that

$$\sum_{i=1}^{n} \frac{1}{i} = \Theta(\log n).$$

(*Hint*: To show an upper bound, decrease each denominator to the next power of two. For a lower bound, increase each denominator to the next power of 2.)

1.9

Starting from the definition of $x \equiv y \bmod N$ (namely, that N divides $x - y$), justify the substitution rule

$$x \equiv x' \bmod N, \quad y \equiv y' \bmod N \quad \Rightarrow \quad x + y \equiv x' + y' \bmod N,$$

and also the corresponding rule for multiplication.

1.10

Show that if $a \equiv b \pmod N$ and if M divides N then $a \equiv b \pmod M$.